

Markscheme

Mathematics: analysis and approaches
Practice question bank

159. (a)

Consider $(1 + i\sqrt{3})^6$.

Use De Moivre's theorem to find its value as a real number.

$$|1 + i\sqrt{3}| = 2, \arg(1 + i\sqrt{3}) = \frac{\pi}{3} \quad (M1) \text{ A1}$$

$$(1 + i\sqrt{3})^6 = 2^6 (\cos 2\pi + i \sin 2\pi) \quad (M1)$$

$$= 64(1) \quad \text{A1}$$

$$= 64$$

[5 marks]